

Activity 1 — Quick Reference

Tools, Extensions, and Commands (VS Code C/C++ Setup)

Tools

Tool	What it is
Visual Studio Code (VS Code)	Free source-code editor from Microsoft; used to write and manage the C/C++ program.
MSYS2	Windows software distribution used to install and manage the MinGW-w64 toolchain.
MinGW-w64	Windows port of GCC; provides the actual C/C++ compiler and debugger.

VS Code Extension

Extension	Purpose
C/C++ (Microsoft)	Adds IntelliSense (code suggestions), syntax highlighting, and debugging support for C/C++ in VS Code.
C/C++ Extension Pack (optional, Microsoft)	Bundles the C/C++ extension with CMake Tools and Code Runner.

Commands

Command	Used for
<code>code --version</code>	Checks that VS Code is installed and accessible from the terminal.
<code>pacman -Syu</code>	Updates the MSYS2 package database and core packages.
<code>pacman -Su</code>	Completes the MSYS2 update after a terminal restart.
<code>pacman -S mingw-w64-ucrt-x86_64-gcc</code>	Installs the GCC compiler toolchain (gcc, g++, gdb) via MSYS2.
<code>gcc --version</code>	Confirms the C compiler is installed and found via PATH.
<code>g++ --version</code>	Confirms the C++ compiler is installed and found via PATH.
<code>gcc hello.c -o hello.exe</code>	Compiles hello.c into an executable named hello.exe.
<code>g++ hello.cpp -o hello.exe</code>	Compiles a C++ source file into an executable (optional step).
<code>.\hello.exe</code>	Runs the compiled executable in the terminal.

Key Concept

Term	Very brief meaning
PATH (environment variable)	Tells Windows which folders to search for command-line programs like gcc, so it can be run from any terminal by name.